



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Introduction to Scalable and Reliable Services

Michele Colajanni

Dipartimento di Informatica – Scienza e Ingegneria
michele.colajanni@unibo.it

Modern services

Google

twitter



facebook

NETFLIX



salesforce

LinkedIn



BitTorrent™



Gmail
by Google

twitch

amazon

YouTube

Apple iTunes

skype™

ebay

Pinterest



Common characteristics of modern services

A **service** is an activity or series of activities of more or less intangible nature that normally take place in interactions between a **service customer** and a **service provider**

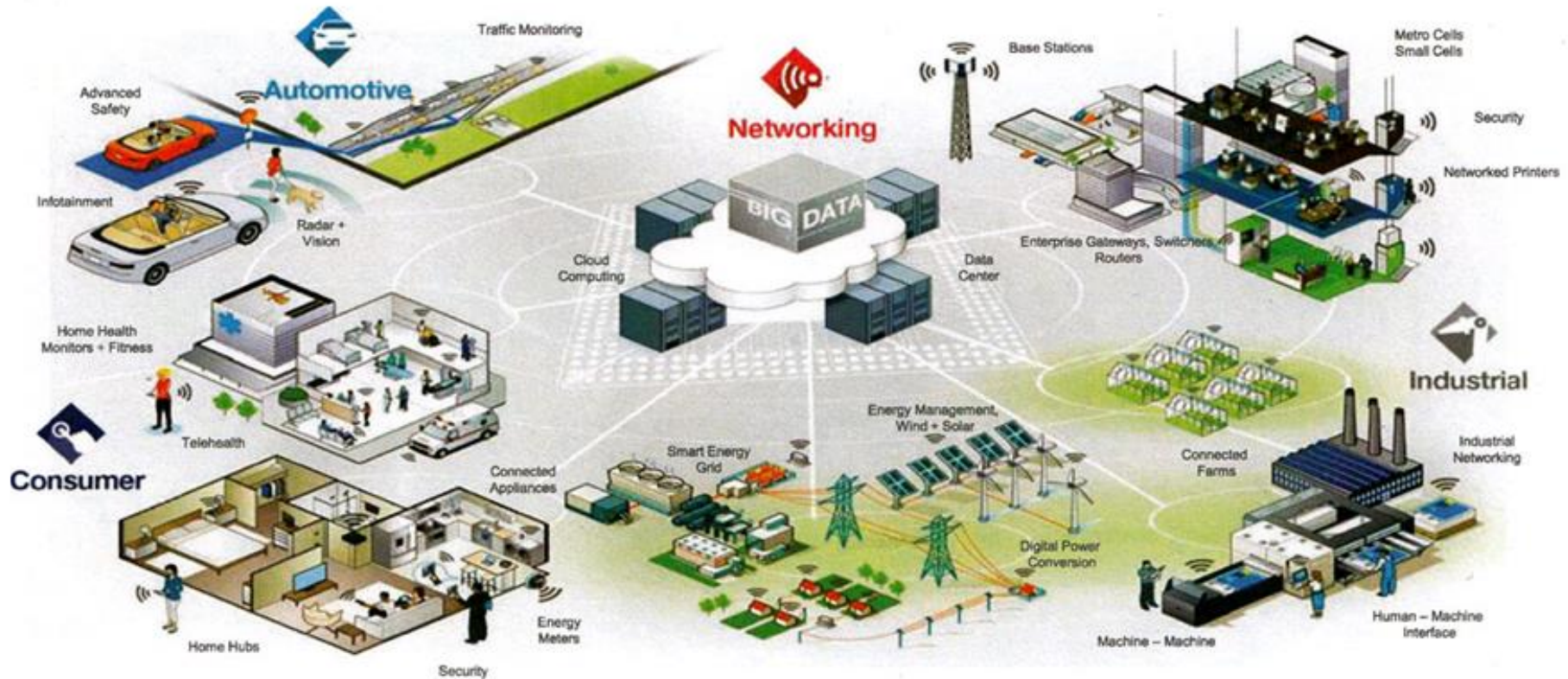
They work always: *resilient* and *secure*

They work well independently of the number of users:
scalable

Their huge architectural complexity is *transparent*



We are building a Cyber Physical world



We are connecting people, data, things, machines, ..., that are enhanced by a lot of **sensors** and **actuators**

We are increasing **self-automation** of smart objects because ...

What's in a *service*?

“A service is the ability of a system (belonging to a *Service provider*) of delivering continuously one or more responses in the presence of specific requests to the system by a customer (*Service requestor*)”

1. There is a **customer** (hopefully, many)
2. There is a **service provider** 
3. There is a **continuity** factor that the service provider must guarantee to the service requestor
4. There is a **quality** factor that is
 - stipulated through a formal contract → **Service Level Agreement (SLA)** including payments and penalties
 - Implicit, but penalties are paid through loss of reputation and trust



From *technology delivery* towards *service delivery*

The alignment of information technology, business methods and organizational practices influence innovation, productivity and client satisfaction in service industries

The fundamental principle behind present and future education models is that they should support the transition from a technology delivery approach towards a service delivery approach based on business goals



Different views

SERVICE

← Customer's view

**Design, Build, Test, Document,
Monitor, Fix, Improve
ARCHITECTURES AND
APPLICATIONS**

← The view in our course



Scale → Complexity

We have a problem of scale of services

**And *scale* changes (and complicates) every well-known
problem in IT research and industry**



A taste of scale: Scenario for the 2020 decade

Billions of Users + Billions of “things”
consuming

Tens of Millions of Services
delivered by

Millions of Service Providers
built on

Hundreds of Millions of Servers
containing

Exabytes of Data
connected by
Terabytes Networks

**Which (HW/SW) technology can
support these applications?**



Exponential times: *information*

- Every two years we generate the same amount of information from the beginning of history to 2003
 - A week of the New York Times contains more information than a person of the 18th century was to come across in an entire lifetime
- ➔ *Human minds and processors have reached their limits: steps ahead require collaboration*



Zero tolerance for unresolved issues

- **Lost prospective customer**
 - If your competitor's service is responsive, you may have lost a sale
 - If the site did not work, or took too long, your customer may not return for a long time – if ever
- **Lost customer**
 - If this happens repeatedly, you have lost a customer
 - AND the customer may stop going to associated physical locations!
- **Lost lives**
 - In a cyberphysical world, safety is an important requirement
- **Lost reputation**
 - People talk about poor performance and reliability; word spreads especially through blogs and social networks



Speed, latency and availability matter

There is substantial evidence suggesting that *performance* and *reliability* directly impacts revenue

- Google reported 20% revenue loss due to a specific experiment that increased the time to display search results by as little as 500 msec
- Amazon reported a 1% sales decrease for an additional delay of as little as 100 msec



A novel cultural approach to the future services

- Reliability is not only a collection of metrics or just a measure for the quality of your infrastructure and software
- It is a big-picture concept based on a collection of practices and cultural lessons that aim to improve service scalability and reliability by holistically incorporating
 - service architects
 - software experts developing and testing code
 - perspective of the users
 - inevitability of errors and failures
 - unavoidable changes and expectations of growth



It is fundamental to understand the impact of reliability metrics, to improve maintenance and incident response, to design and code with a focus on service level objectives and error budgets



Services and business goals

- For each metric, organizations need to make decisions on how much time loss and frequency of failures they can bear without disrupting the overall system performance for end-user
- Similarly, they need to decide how much they can afford to spend on the service, infrastructure and support to meet certain standards of availability and reliability of the system
- The resulting strategy is a trade-off between **cost** and **service levels** in context of the impact and requirements for maintaining a reliable and available service. Two main categories:
 - **Business value**
 - **Critical infrastructures**



Business cases with a strong commitment on reliability

Datacenters are so crucial to e-services that the companies have determined exactly how much they lose whenever there is an outage

Mazen Rawashdeh (*eBay*): **“Depending on what goes down and when, the impact on the site is from \$2,000 to \$4,000 per second” (120,000 – 240,000€ per minute)**

For Amazon is about 150,000\$ per minute

For Facebook is about 120,000\$ per minute



Disturbances



INTENTIONAL ACTION

External attacks

Supply chain attacks

Internal attacks

NEGLIGENT ACTION

Human errors



INTERNAL ACCIDENTAL EVENT

Failures

Incidents

EXTERNAL ACCIDENTAL EVENT

Natural disasters

Changes in Law

SUPPLIER ACCIDENTAL EVENT

Failures

Incidents

Changes in market rules

Technological innovations



Disturbances



INTENTIONAL ACTION

External attacks

Supply chain attacks

Internal attacks

NEGLIGENT ACTION

Human errors



INTERNAL ACCIDENTAL EVENT

Failures

Incidents

EXTERNAL ACCIDENTAL EVENT

Natural disasters

Changes in Law

SUPPLIER ACCIDENTAL EVENT

Failures

Incidents

Changes in market rules

Technological innovations



Accidental threat types

1. Human adversaries

- Intentional
- Negligent

2. Internal failures and incident

- Black out
- Network and/or system outage or malfunction
- Water leak
- Short circuit and fire

3. External natural events

- Flooding, extreme weather, earthquakes, ...

4. External human events

- Market and law changes
- Technology innovations
- Diseases, pandemics, strikes, riots, ...



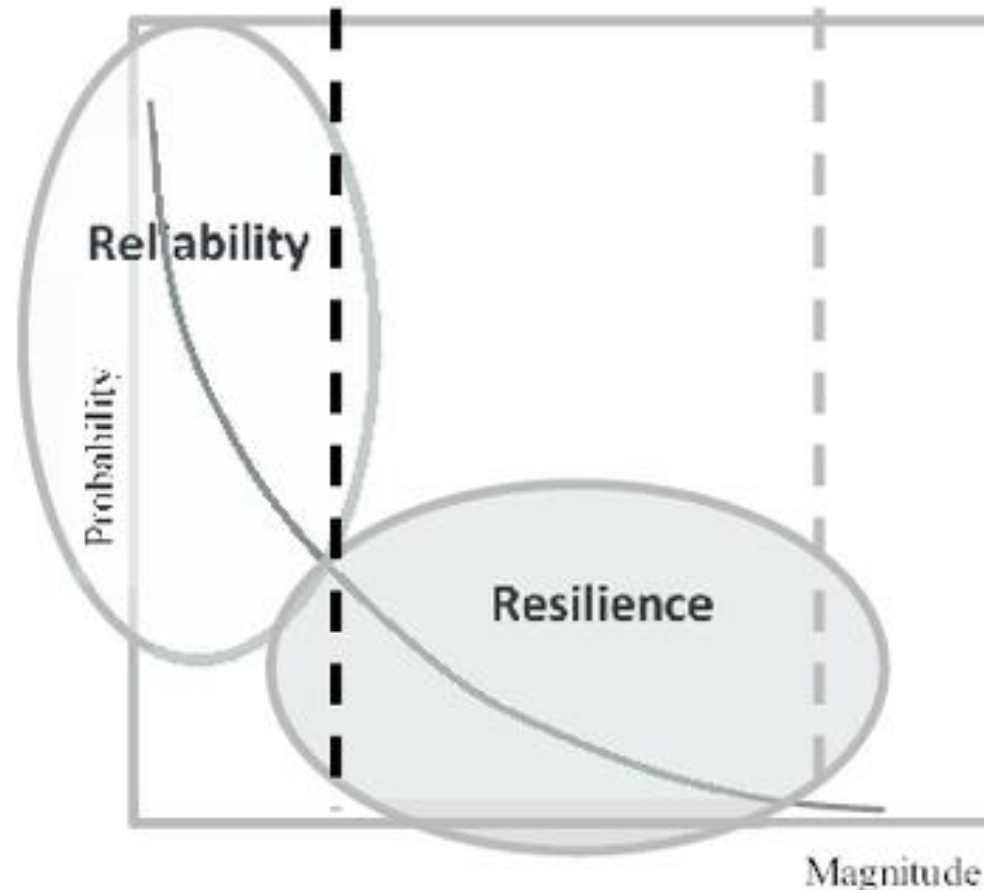
Reliability and Resilience

- **Reliability** is usually measured in relation to expected or regulated levels of service
- **Resilience** is the property and/or performance of the system when the level of service has not been attained for whatever reason, but particularly when subject to unexpected or exceptional disturbances

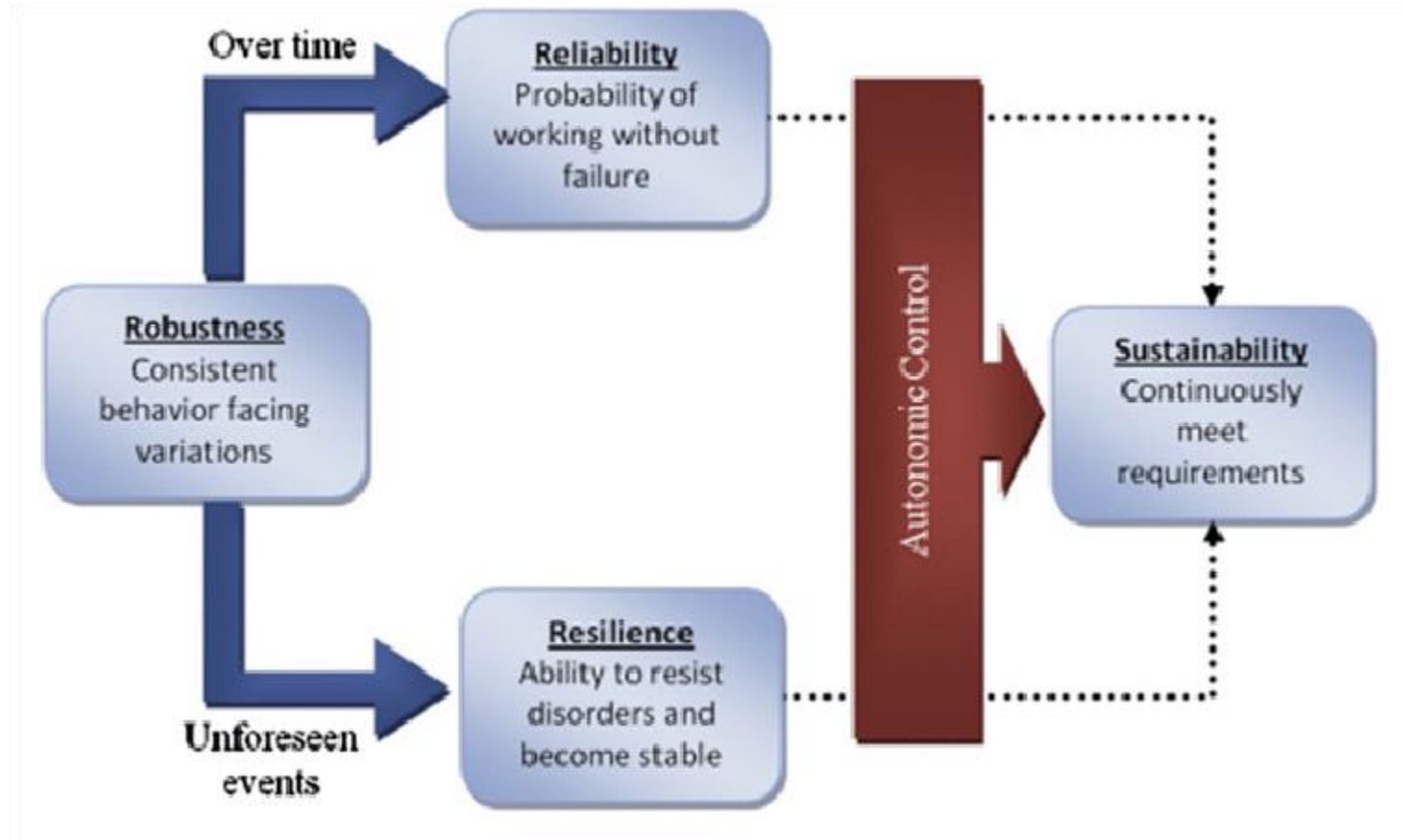


Reliability and Resilience are related to *risk management*

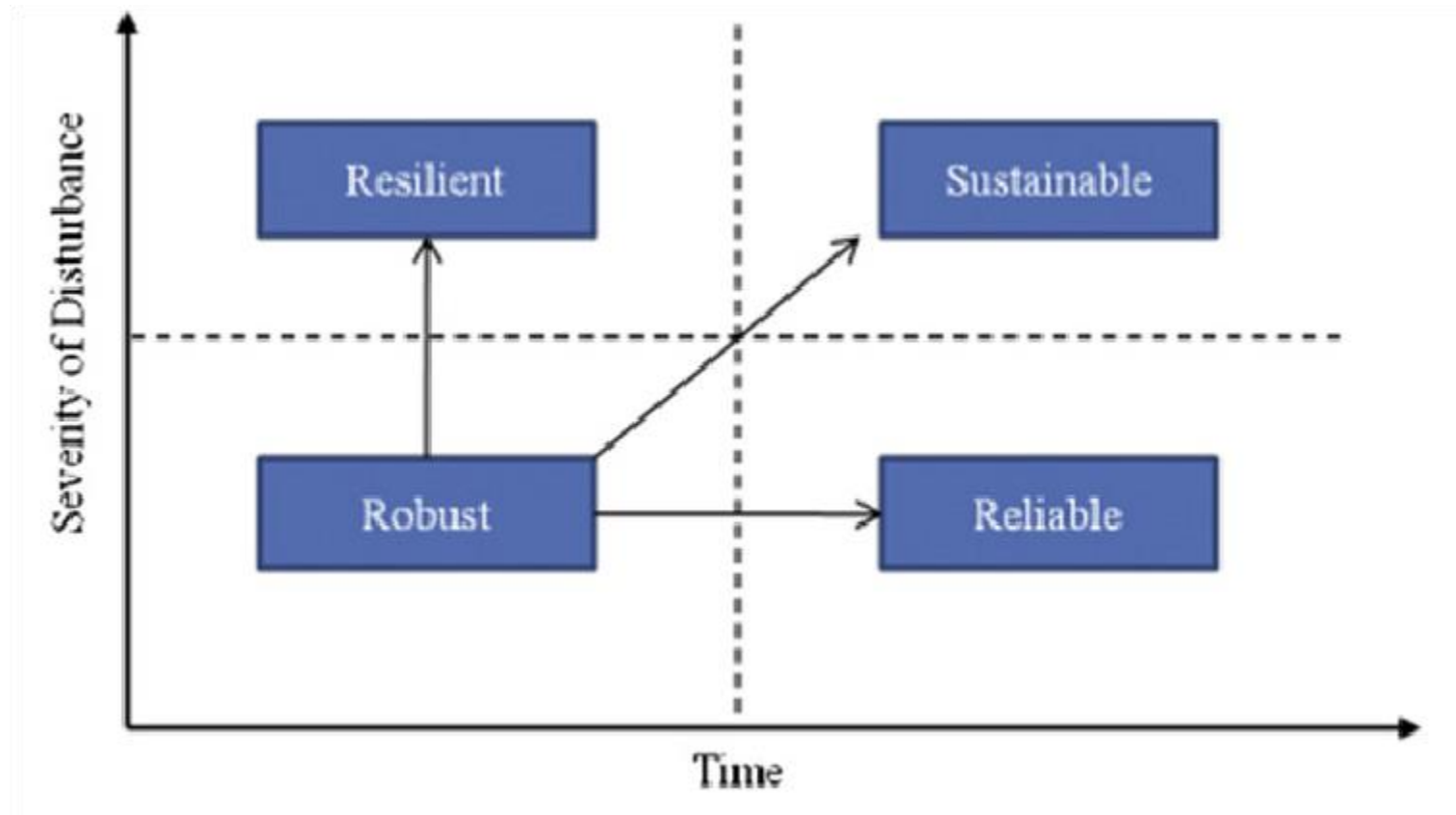
How reliability and resilience are related with respect to the **probability** and **magnitude** of a *disturbance*



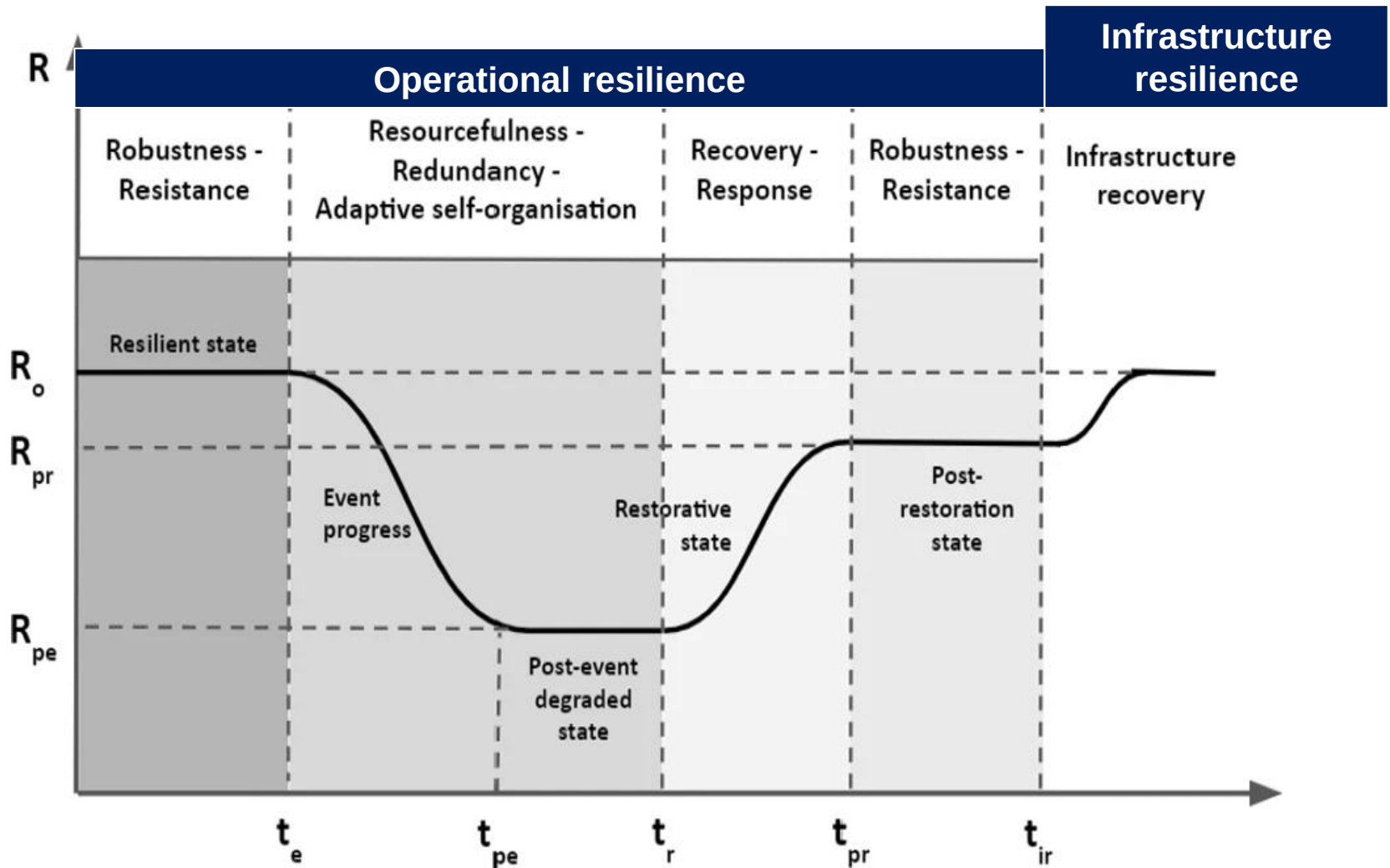
Risks and Countermeasures



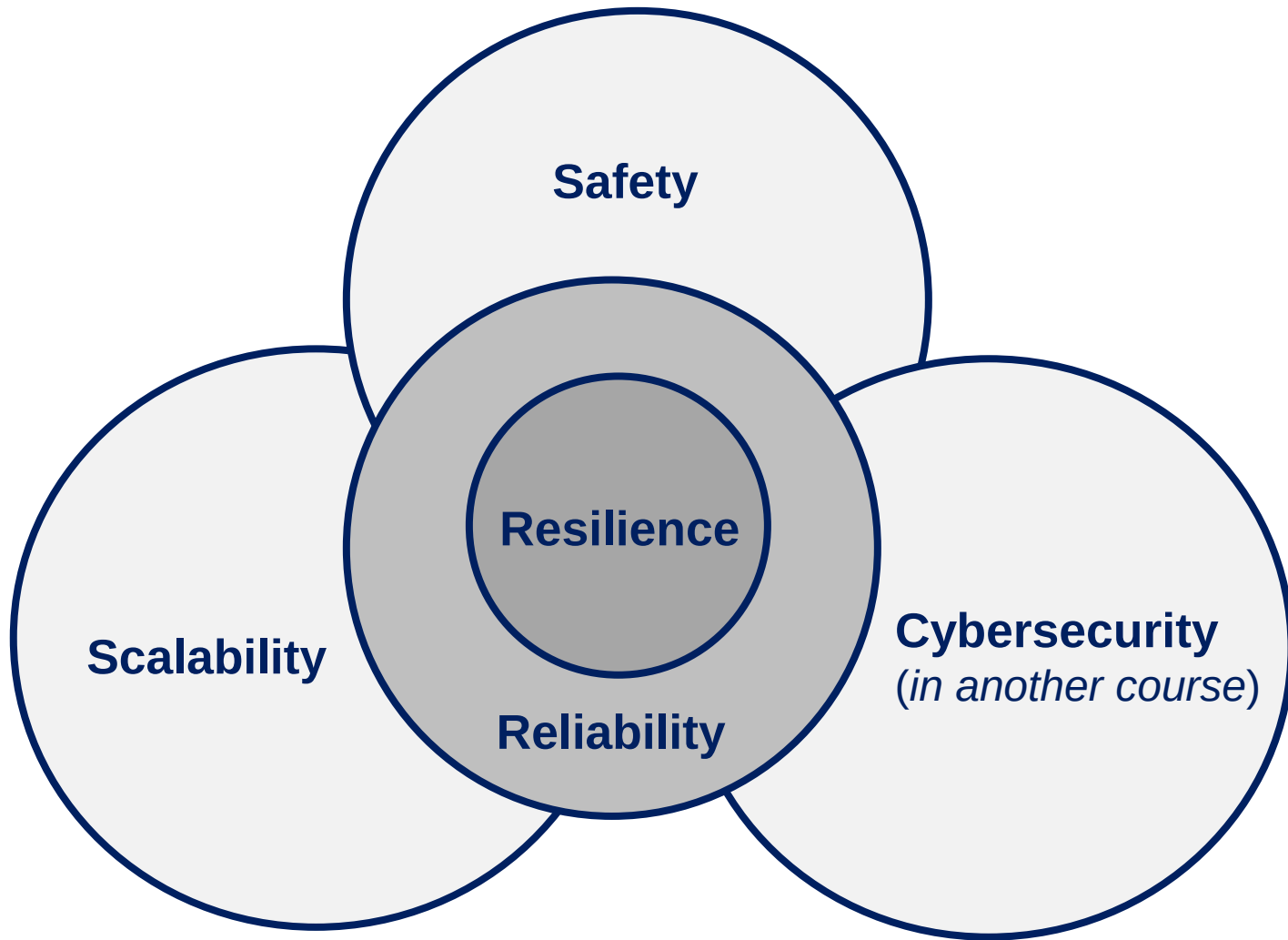
Severe and prolonged *disturbance*



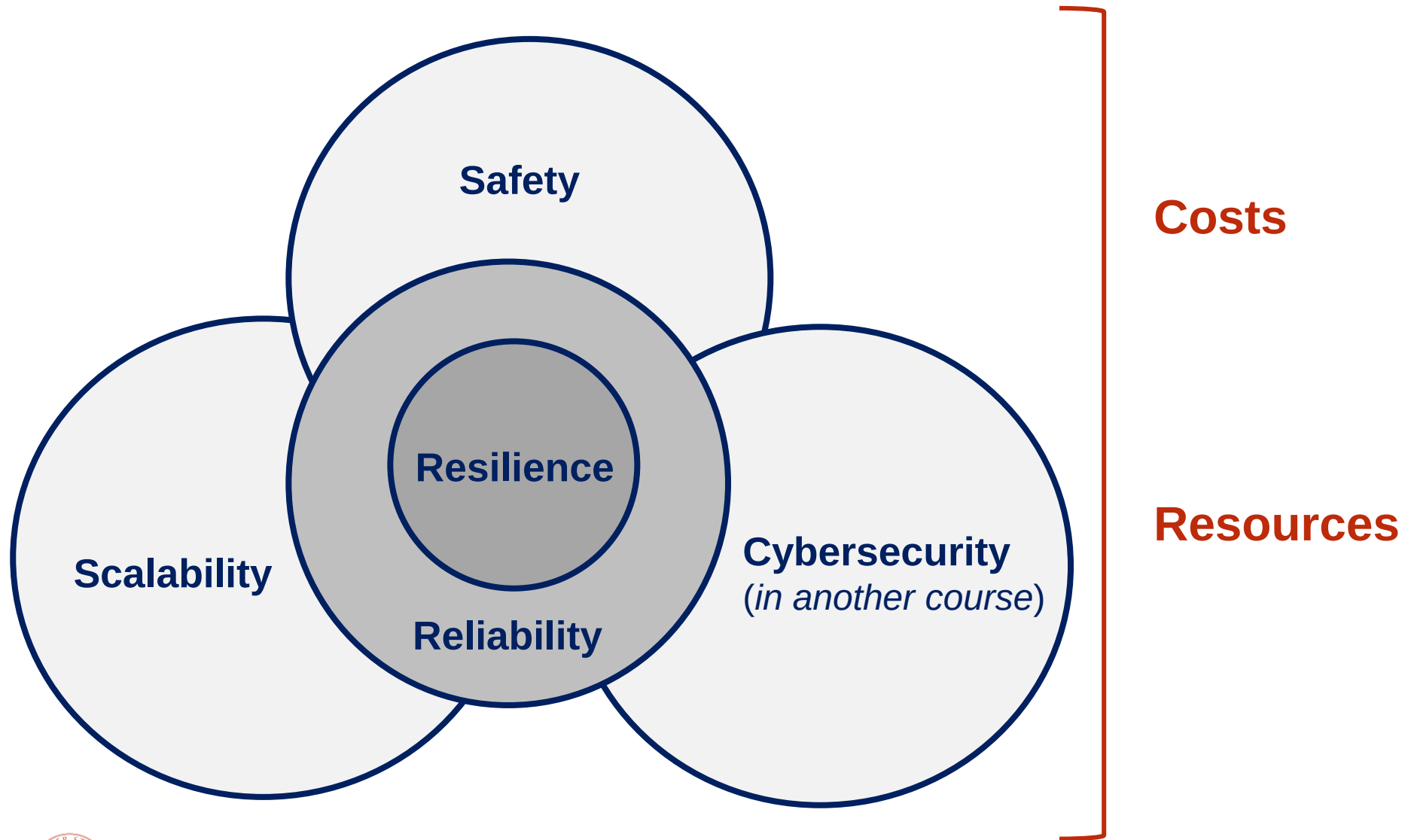
The overall vision



Design principles of modern services



Modern services and constraints of reality



Learned lesson #1

Perfect solution → Unlimited budget

Engineering is not about perfect solutions.

It's about doing the best you can with limited resources.

[Randy Pausch]



Any scalable and reliable architecture must avoid

System bottlenecks

Single points of failure

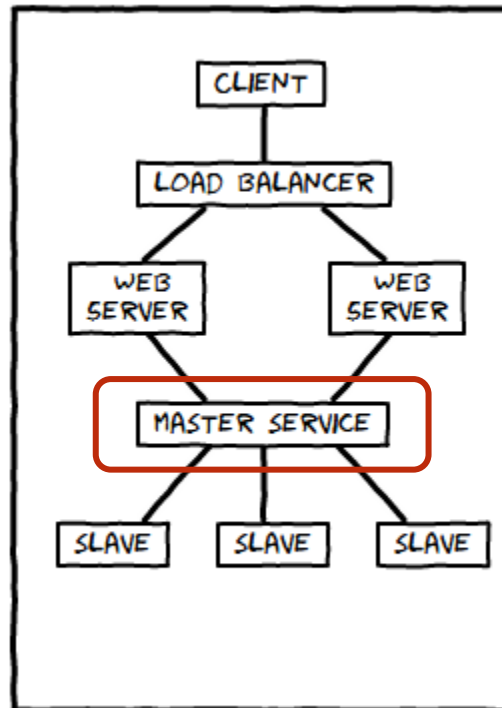


Know your Enemy

ONE CRAZY MORNING

MAN, THE SERVERS
CAME DOWN YET
AGAIN TODAY

DUDE, AGAIN? WHAT'S THE
ARCHITECTURE
LOOK LIKE?



WHAT'S THE MASTER SERVICE?
IT LOOKS LIKE A BOTTLENECK

IT'S LIKE A LOAD
BALANCER BUT
IT DOES MORE.

WELL THAT'S YOUR
PROBLEM, THE MASTER
OVERLOADS AND THE
WHOLE CLUSTER
CRAPS OUT

YEP...



Best solution to address *scalable* and *reliability* issues?

Replication, Replication, Replication

Replication of server machines

Replication of interconnections

Replication of data

Replication of application processes

Replication of people

...



SRS course

ARCHITECTURE

APPLICATION

